Banach Fixed Point Theorem

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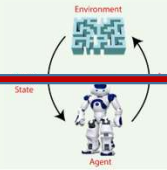
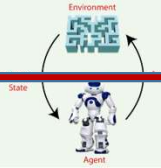


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Overview

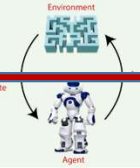
The claim

For each MDP, there is always at least one policy that is better than or equal to all other policies.

The claim is proved with the following theorem

Theorem: *For any finite MDP, there exists an optimal policy π^* such that it is better than or equal to every other possible policy π.*

When is one policy (π_1) considered better than the other policy (π_2)?

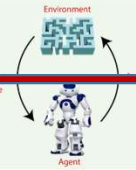


Overview

- If the value of a state derived using π_1 is better than or equal the value of a state derived using π_2 for every state in the environment, then the policy π_1 is said to be better than policy π_2.

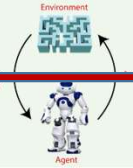
$$\pi_1 \geq \pi_2 \text{ if } V_{\pi_1}(s) \geq V_{\pi_2}(s) \forall s \in S$$

- Now that we know how to compare policies, we need to prove that there always exists a policy that is better than all other policies.



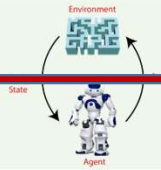
Overview

- We are going to prove this using the **Banach fixed point theorem** by showing that the **Bellman optimality operator** is a **contraction** over a **complete metric space** of real numbers with metric **L-infinity norm**.
- For this, we will first discuss the **fixed point problem** and **complete metric spaces** with respect to the **Cauchy sequence**.



Fixed point problem

- Usually we solve for x such that a function $f(x) = 0$.
- If we solve for x such that $f(x) = x$ → fixed point problem
- As the name suggests, x is a fixed point, it does not change even on the application of the function.
- A fixed point problem can be converted into a problem of finding roots by forming another function $g(x) = f(x) - x = 0$.

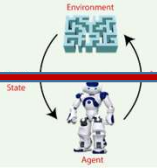


Fixed point problem

- *To solve a fixed point problem, choose a random starting value of x and repeatedly apply f(x) infinite times. If the* ***function is convergent*** *and you're lucky, you will find the solution.*
- Mathematically

$$f^n(x) : \text{apply the function f() n times on x}$$
$$\text{example, } f^2(x) = f(f(x))$$

- Now, if the function is convergent then it must converge to some value, say, x^*. This value, x^* is indeed the solution to the fixed point problem



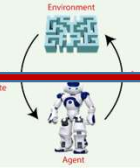
Fixed point problem

- Let's choose some arbitrary value x_0 and apply the function f(.) on x_0 for infinite times to get x^*, and then use that to solve the fixed point problem:
- The intuition behind this is pretty simple, if a function has converged at some point then the value of this function at that convergent point will be the convergent point itself

$$
\begin{aligned}
\text{let}, x^* &= \lim_{n\to\infty} f^n(x_o) \\
f(x^*) &= f(\lim_{n\to\infty} f^n(x_o)) \\
&= \lim_{n\to\infty} f(f^n(x_o)) \\
&= \lim_{n\to\infty} f^{n+1}(x_o) \\
&= x^*
\end{aligned}
$$

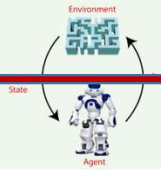
Metric space

- A metric space is simply a set with a metric defined to measure the distance between any two elements of the set.
- For example, the Euclidean space is a metric space with distance defined as the Euclidean distance in the set of real numbers.
- A metric space M is represented as (X, d) where X is the set and d is some sort of metric.
- Metric d can be something that can be used to measure the size of the vector in the defined space → Norm
- There are certain properties that the metric d must satisfy.



Metric space

- Identify $d(x, x) = 0$
- Non-Negativity $d(x, y) > 0$
- Symmetry: $d(x, y) = d(y, x)$
- Triangular inequality: $d(x, z) \leq d(x, y) + d(y, z)$



Cauchy sequence

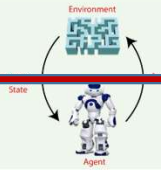
- For a metric space (X, d) the sequence of elements of the set X, (x1, x2, x3…. xn) is a Cauchy sequence if, for every positive real number ε, there exists an integer N such that the following equation holds:

$$d(x_a, x_b) < \varepsilon; \;\; a, b > N$$

- A sequence of elements of a metric space is Cauchy if this sequence converges at some point (the distance between them becomes constant).

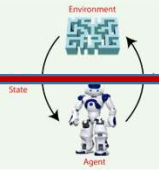
Cauchy sequence

- Let Cauchy sequence be x1, x2, x3……. For any ϵ, we can find some point in sequence beyond which if we pick two elements, they are within ϵ of each other.
- For any small distance, there is a certain index past which any two terms are within that distance of each other, which captures the intuitive idea of the terms becoming close



Complete Metric space

- A metric space (X, d) is complete if every possible Cauchy sequence of the elements in the set X converges to an element that also belongs to the set X.
- That is to say, the convergent limit of every Cauchy sequence of the elements of set lies in the set itself. Which is why it's called "complete".

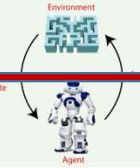


Contraction

- A function (or operator or mapping) defined on the elements of the metric space (X, d) is a contraction (or contractor) if there exists some constant $\gamma \in [0,1)$ such that for any two elements of the metric space x1 and x2, the following condition holds:

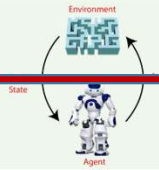
$$d(f(x_1), f(x_2)) \leq \gamma d(x_1, x_2)$$

- This means that after applying the mapping f(.) on the elements x1 and x2, they got closer to each other by at least a factor γ.



Contraction

- The smallest value of such a constant γ is called the Lipschitz constant (this is an important constant for generative adversarial networks).
- If γ=1, the mapping is no more a contraction but rather a short mapping.
- Intuitively, it can be observed that the sequential values of the elements are getting closer after applying the contraction mapping.



Contraction

- Take X=[0,2] with the usual distance and define $f(x)=(x+3)/4$.

1. Show $f:X\rightarrow X$
2. Show f is a contraction and find k
3. Find the fixed point
4. Iterate from starting point and show convergence

Contraction

- Take X=[0,2] with the usual distance and define $f(x)=(x+3)/4$.

1. Show $f: X \to X$
2. Show f is a contraction and find k

If $x \in [0,2]$, then $f(x) = (x+3)/4 \in [(0+3)/4, (2+3)/4] = [0.75, 1.25] \subset [0,2]$. So f maps X into itself.

For any x, y,

$$|f(x) - f(y)| = \left| \frac{x+3}{4} - \frac{y+3}{4} \right| = \frac{1}{4}|x-y|.$$

Contraction

- Take X=[0,2] with the usual distance and define $f(x)=(x+3)/4$.

1. Find the fixed point
2. Iterate from starting point and show convergence

Solve $x = (x+3)/4$:

$$4x = x + 3 \implies 3x = 3 \implies x^* = 1.$$

Iterate from a starting point (say $x_0 = 0$) and show convergence.

Define $x_{n+1} = f(x_n)$. Compute a few terms:

$$x_0 = 0,$$
$$x_1 = f(0) = 0.75,$$
$$x_2 = f(0.75) = 0.9375,$$
$$x_3 = 0.984375,$$
$$x_4 = 0.99609375,$$

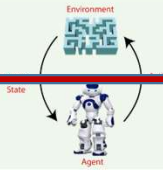
Contraction

Iterate from starting point and show convergence

Iterate from a starting point (say $x_0 = 0$) and show convergence.

Define $x_{n+1} = f(x_n)$. Compute a few terms:

$$x_0 = 0,$$
$$x_1 = f(0) = 0.75,$$
$$x_2 = f(0.75) = 0.9375,$$
$$x_3 = 0.984375,$$
$$x_4 = 0.99609375,$$
$$x_5 = 0.9990234375, \ldots$$



Questions in previous exams

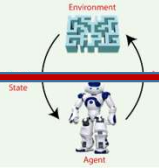
- Show that $f(x) = 0.5 \sin x$ is a contraction mapping by showing convergence starting from $x_0 = 0.2$
- Show that $f(x) = 0.25 \cos^2 x$ is a contraction mapping by showing convergence starting from $x_0 = 0.2$

Banach Fixed point Theorem

- Informally, this theorem says that for a complete metric space, the application of a contractor on the elements of the set, again and again, would eventually get us to an optimal, unique value.

We know:

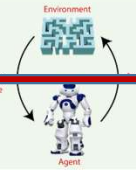
- Contractors bring the elements of the set together.
- Applying this contractor, again and again, would get us a sequence. (Cauchy?)
- The Cauchy sequences in a complete metric space always converge to a value that is part of the metric space.



Proof of Uniqueness

Theorem: Let (X, d) be a complete metric space and a function f: X->X be a contractor then, f has a ***unique*** *fixed point x*∈X (i.e. f(x*)=x*) such that the sequence f(f(f(...f(x)))* ***converges*** *to x*.*

To prove this mathematically, we need to prove both the uniqueness and existence of x*



Proof of Uniqueness

- Uniqueness: We will prove this by contradiction. Let us assume that the convergence value is not unique and x1* and x2* are two values at which the sequence of contractor converges then, we will have:

$$d(f(x_1^*), f(x_2^*)) = d(x_1^*, x_2^*) \quad (1)$$

x1* and x2* are optimal values, the contractor has converged here, the distance does not change anymore

- Also, note that f is a contractor so it must hold the following property:

$$d(f(x_1^*), f(x_2^*)) \leq \gamma d(x_1^*, x_2^*) \quad (2)$$

The contractor property

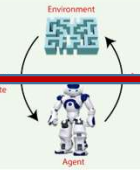
- Now since $\gamma \in [0,1)$, it's impossible to satisfy both equation 1 and 2 simultaneously. Therefore our assumption must be wrong. Hence, by contradiction, x* must be unique.

Proof of existence

- Now that we've proved that x* is unique, we need to prove that x* exists. Let (x1, x2, x3, …. xn) be the sequence formed by repeatedly applying the contractor.

$$x_n = f^n(x)$$

- If we assume that the sequence (x_1, x_2, x_3, …. x_n) is Cauchy, we know for certain that this sequence will converge to some point, say, x*. Also, since the metric space is complete, this convergent point, x* will belong to the metric space (X, d)
- we just need to prove that this sequence is Cauchy.



Proof that the sequence is Cauchy

- Assume two elements of the set x_n and x_m such that $m>>n$ and, m is very large, then by repeatedly applying the triangular inequality property of the metric d, we have:

$$
\begin{aligned}
d(x_m, x_n) &\leq d(x_m, x_{m-1}) + d(x_{m-1}, x_n) \\
&\leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + d(x_{m-2}, x_n) \\
&\leq d(x_m, x_{m-1}) + \ldots. + d(x_{n+1}, x_n) \\
&\leq d(f^m(x_o), f^{m-1}(x_o)) + \ldots. + d(f^{n+1}(x_o), f^n(x_o))
\end{aligned}
$$

Expanding the triangular inequality of the metric d

Proof that the sequence is Cauchys

- Now, since f is a contractor, we know that:

$$d(f^n(x_1), f^n(x_2)) \leq \gamma d(f^{n-1}(x_1), f^{n-1}(x_2)).... \leq \gamma^{n-1} d(f(x_1), f(x_2)) \leq \gamma^n d(x_1, x_2)$$

- We can further reduce $d(x_m, x_n)$ as follows:

$$\begin{aligned} d(x_m, x_n) &\leq d(f^m(x_o), f^{m-1}(x_o)) + ... + d(f^{n+1}(x_o), f^n(x_o)) \\ &\leq \gamma^{m-1} d(f(x_o), x_o) + ... + \gamma^n d(f(x_o), x_o) \\ &\leq d(f(x_o), x_o) \sum_{i=n}^{m-1} \gamma^i \end{aligned}$$

Proof that the sequence is Cauchys

$$\leq \gamma^n d(f(x_o), x_o) \sum_{i=0}^{m-n-1} \gamma^i$$

$$\leq \gamma^n d(f(x_o), x_o) \sum_{i=0}^{\infty} \gamma^i \quad \text{(m-n-1 is very large)}$$

$$\leq \frac{\gamma^n}{(1-\gamma)} d(f(x_o), x_o)$$

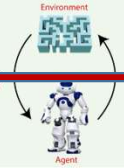
- Now, by choosing n to be sufficiently large, we can make the RHS of the above equation less than any positive real number ε. Hence, the sequence (x1, x2, x3, …. xn) is Cauchy and an optimal $x^* \in X$ exists.

Bellman Optimality Equation

- For the value function, V(s) we define a new operator, the optimal Bellman operator, B which takes in a value function and returns another value function.

$$BV(s) = \max_a R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a V(s')$$

- It can easily be observed, *B* is a recursive operator. Therefore, this will generate a sequence of value functions.
- If we can show that *B* is indeed a contractor for some metric space (*X*, *d*) then by the Banach fixed point theorem, we can conclude that the repeated application of the optimal Bellman operator will eventually give a unique optimal value function using which an optimal (best) policy can be derived.



Bellman Optimality Equation

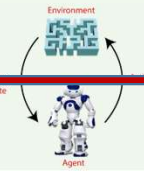
- Therefore, all our work now reduces to proving that B is a contractor.
- Metric space (X,d): The set X is the set of real numbers defined as follows:

$$X : V(s) \in \mathbb{R}$$

$$X : V \in \mathbb{R}^{|\mathbb{S}|}$$

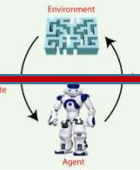
- For the metric, we use the L-infinity norm defined as follows:

$$||X||_\infty = \max_{i \in [0,|X|]} |X_i|$$



Bellman Optimality Equation

- According to L_∞ norm, the distance between the two value functions will be equal to the highest element-wise absolute difference between the two.
- For finite MDPs with finite rewards, the value functions will always stay in the real space.
- It is impossible for the value function to not be in the real space, therefore, this finite space will always be complete.

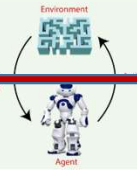


Bellman Optimality Equation

Theorem: Bellman operator B is a contraction mapping in the finite space (R, L-infinity)

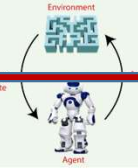
Proof: Let V1 and V2 be two value functions. Then:

$$
\begin{aligned}
||BV_1(s) - BV_2(s)|| &= ||\max_a (R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a V_1(s')) - \max_{a'} (R_s^{a'} + \gamma \sum_{s' \in S} P_{ss'}^{a'} V_2(s'))|| \\
&\leq ||\max_a (R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a V_1(s') - R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a V_2(s'))|| \\
&\leq \gamma ||\max_a (\sum_{s' \in S} P_{ss'}^a (V_1(s') - V_2(s'))|| \\
&\leq \gamma \max_{a,s'} \sum_{s' \in S} P_{ss'}^a (V_1(s') - V_2(s')) \\
&\leq \gamma ||V_1(s') - V_2(s')|| \max_{a,s'} \sum_{s' \in S} P_{ss'}^a \\
&\leq \gamma ||V_1(s') - V_2(s')||
\end{aligned}
$$



Bellman Optimality

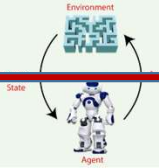
- In the second step above, we introduce inequality by replacing *a'* by *a* for the second value function. This is because by replacing its optimal action *a'* by some other action *a*, we have reduced its overall value thereby introducing an inequality.
- In the fourth step, we remove the L-infinity norm by taking the max over s' (recall the definition of L-infinity with respect to value functions in our setting)
- In the final step, we remove the sigma because the sum of probabilities is always 1.



Bellman Optimality

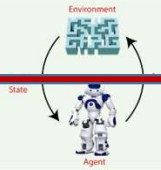
Now we know:

- (R, L-infinity) is a complete metric space
- Bellman operator B is a contractor
- Hence, by the Banach fixed point theorem, we conclude that there exists a unique optimal value function V* for every MDP. Using this V*, we can derive the optimal policy π^*.
- Hence proved, for any finite MDP, there exists an optimal policy π^* such that it is better than or equal to every other possible policy π.



What next??

- Now, how to find this optimal policy and value function?
- One way is to just repeatedly apply the Bellman operator to a random initial value function to get the optimal function. But, this is computationally very expensive and often downright infeasible.
- Use iterative methods like value and policy iteration or temporal difference methods like Q-Learning or SARSA.



Reference

[Mathematical Analysis of Reinforcement Learning — Bellman Optimality Equation | by Vaibhav Kumar | Towards Data Science]

